CSC-236 - Computer Organization & Assembly Language

Roll no:	**Name:**	Instructor: Sheeza Batool
Assignmnet-1		Max Marks: **20 marks**
Date : 6th Mar, 2026		Due Date : 10th Mar, 2026

Computer Organization and Assembly Language (COAL)

Assignment 1

Assignment 1 contains questions related to 8086 architecture, effective address calculation, arithmetic operations, and status/control flags. Show all calculation steps where required.

Section A: Effective Address Calculations

1. The value of Code Segment (CS) Register is 4042H and the value of different offsets is as follows:
   BX: 2025H , IP: 0580H , DI: 4247H
   Calculate the physical address of the memory location pointed by the CS register with each offset.
2. Calculate the physical address for the following register values:
   SS: 3860H, SP: 1735H, BP: 4826H
3. The value of the DS register is 3032H. BX = 3032H. Execute the instruction ADD BX, 0008H.
   AX contains a value which is stored using MOV [BX], AX.
   Calculate the physical memory address where AX will be stored.
4. You are provided the following values:
   DS: 3056H, IP: 1023H, BP: 2322H and SP: 3029H
   Calculate the physical address using DS with BP and SP.
5. CS = 2A40H, IP = 1F20H. Calculate the physical address of the next instruction.
6. DS = 4500H, SI = 1200H. Calculate the physical address accessed by DS:SI.
7. ES = 3890H, DI = 2000H. Calculate the physical address accessed by ES:DI.
8. The Extra Segment (ES) register contains the value 3B20H. The DI register initially contains the value 0200H. An instruction is executed : ADD DI, 0015H
   The BX register contains a value that must be stored in memory using the instruction: MOV ES:[DI], BX
   Calculate the physical memory address where the value of BX will be stored.

Section B: Arithmetic and logical Operations and Status Flags

Answer Table Example for all Arithmetic and logical Questions

Operation	**Result**	**Sign**	**Zero**	**Carry**	**Overflow**	**Aux Carry**	**Parity**
ADD	**34ADH**	**1**	**0**	**0**	**1**	**0**	**1**

Determine the values of Sign, Zero, Carry, Overflow, Auxiliary Carry and Parity flags.

9. Perform: 4A3FH + 8B21H.
10. Perform: 7F20H + 0105H.
11. Perform: A210H - 3B2FH.
12. Perform: 9C45H + 6A2BH.
13. Perform: 3A2FH - 4B10H.

14. Perform: F234H + 1200H.
15. Perform: 5A10H AND 3F22H.
16. Perform: 8A2FH OR 1204H.
17. Perform: 7A3CH XOR 1F0FH.

Section C: Control Flags

18. Explain the purpose of the following control flags in the 8086 microprocessor: Trap Flag (TF), Interrupt Flag (IF), Direction Flag (DF).

Section D: Architecture Comparison

19. Compare the architecture of the Intel 8086 and Intel 8088 microprocessors. Explain at least five differences including data bus width, instruction queue, and system design.

20. Draw and explain the block diagram of the 8086 microprocessor architecture.